

Ramy Lazghab

AI & Machine Learning Engineer

LLMs • Agentic AI • RAG • Production ML Systems

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LinkedIn | GitHub | Portfolio

Core Technical Skills

Programming: Python, SQL, Java, JavaScript, C, R

Machine Learning: Deep Learning, Reinforcement Learning, NLP (Natural Language Processing), Transformers, Fine-tuning, Gradient Boosting, Model Explainability (SHAP); PyTorch, TensorFlow, XGBoost, LightGBM

Generative AI & LLM: RAG, Multi-Agent Systems, LLM Evaluation, Prompt Engineering; LangChain, LangGraph, Hugging Face, Anthropic API, OpenAI API, MCP, LangSmith

MLOps, Cloud & Data: Docker, Git, MLflow, GCP (Vertex AI, BigQuery, Cloud Run), REST APIs, PostgreSQL, Neo4j, vector databases (FAISS, Qdrant), Spark, Pandas, NumPy, Streamlit

Professional Experience

AI Engineer Intern

Apr 2026 – Present

Relay X, Paris

- **Oracle AI, regulatory risk forecasting:** predicts whether a regulator will restrict an ingredient within 6 or 18 months. 1,222 scientific opinions into a 70-feature vector; XGBoost + LightGBM ensemble, isotonic calibration, SHAP explanations. **AUC-ROC 0.948 at the 6-month horizon and 0.914 at 18 months**, under strict temporal validation on held-out opinions.
- Live enrichment through a LangGraph pipeline (Anthropic Claude) with three parallel fetch nodes (PubMed, PubChem, web), rendered as a Streamlit intelligence report; LangSmith tracing.
- **PIF (dossier generation):** six-agent LLM pipeline producing EU Product Information Files from client documents, with template-driven sections and human-in-the-loop safety validation. **Shipped into the production platform** (Next.js, TypeScript, Express, Prisma, PostgreSQL).
- Authored a golden-case benchmark (5 products, 203 ground-truth fields) and used it as a correction loop across three pipeline versions: **unsupported claims cut from 87.5% to 35.0%, F1 raised from 81.8 to 89.5, at \$0.13 per dossier.**
- **Magnet (regulatory radar):** five-tier search cascade (SQL first, LLM only on a miss), three-pass source-cited reporting. Benchmarked across 35 generated reports at **88.6% source provenance**; 75 API security tests.

Software Developer Intern

Feb – Jun 2024

Infinity Management

E-learning platform using NLP and BERT-based sentiment analysis to personalize learning; Django REST, React, Docker.

Education

M.Sc. Data Science & Artificial Intelligence

2024 – 2026 (expected)

Université Paris Dauphine – PSL

B.Sc. Computer Engineering (IoT)

2021 – 2024

Faculty of Sciences of Tunis

Selected Projects

- **TelecomPlus:** multi-agent customer support (RAG + SQL routing), LLM-as-a-Judge evaluation, Langfuse & MLflow tracking. *LangChain, LangGraph, FAISS, PostgreSQL*
- **RAGenius:** AI research assistant answering questions over PDF and CSV via semantic retrieval. *LangChain, Hugging Face, FAISS*
- **Movie Recommender:** end-to-end recommendation system on GCP, exposed as a REST API. *BigQuery ML, Vertex AI, Cloud Run*
- **Alzheimer's Prediction:** ensemble of XGBoost and deep learning for early diagnosis; SHAP for clinical interpretability. *PyTorch, XGBoost, SHAP*

Certifications & Achievements

- **Certifications:** LangChain Academy ; Google Cloud, BigQuery ML
- **3rd Place**, RAISE Summit AI Hackathon Paris ; **Winner**, Hack for Good (MoodSync) ; EY Hack for Smart Insurance ; IEEE Xtreme 15.0/16.0 ; IEEE, ITLAB & AIESEC member
- **Languages:** English, fluent (IELTS) | French, professional